



GLA UNIVERSITY, MATHURA

Department of Computer Engineering Application

Vehicle Rental & Fleet Management System

SYNOPSIS

for

B.Tech CSE 3rd Year

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Abstract

The Vehicle Rental & Fleet Management System is a frontend-based web application designed to provide a simple and user-friendly platform for vehicle rental and fleet management. The system allows users to browse available vehicles, view vehicle details, select rental dates, and make a demo booking. It also provides an administrative interface for managing vehicle information, booking records, availability, and maintenance details.

The project is developed using HTML, CSS, and JavaScript. HTML provides the structure of the application, CSS is used to create a responsive and attractive user interface, and JavaScript provides interactive features such as vehicle searching, filtering, booking calculations, form validation, and dynamic status updates. Sample data and browser LocalStorage can be used to demonstrate the system without a backend database.

The expected outcome is a responsive and interactive frontend application that demonstrates the basic workflow of a modern vehicle rental and fleet management platform.

Keywords

HTML, CSS, JavaScript, Vehicle Rental,
Fleet Management, Web Application,
Responsive Design, LocalStorage

1. Introduction

The vehicle rental industry requires an efficient way to present vehicle information, manage availability, and handle rental bookings. Traditional manual methods of maintaining vehicle and booking information can be time-consuming and difficult to organize.

The proposed Vehicle Rental & Fleet Management System provides a web-based frontend interface where customers can explore available vehicles, check their details, select rental dates, and make demo bookings. The system also includes an admin dashboard through which vehicle, booking, customer, and maintenance information can be displayed and managed. The project focuses on frontend development using HTML, CSS, and JavaScript, with an emphasis on responsive design, user interaction, and easy navigation.

2. Problem Statement

Existing manual or partially digitized rental workflows make it difficult to determine vehicle availability in real time, often resulting in double bookings, delayed responses to customer requests, and inconsistent rental pricing. Fleet-related information such as vehicle condition, servicing history, and maintenance schedules is frequently maintained separately from booking records, increasing the risk of assigning a vehicle that is due for maintenance.

There is a need for an integrated system that allows customers to independently search and book vehicles online while giving administrators a consolidated view of the fleet, active bookings, and maintenance status, thereby reducing operational errors and improving service reliability.

3. Objectives

- To design and develop a centralized web-based platform for managing vehicle rentals and fleet operations.
- To allow registered users to search, view, and check the real-time availability of vehicles before booking.

- To enable users to book and cancel vehicle rentals through a simple, guided online process.
- To automate rental price calculation based on rental duration and vehicle category.
- To provide administrators with tools to manage the vehicle fleet, customer records, and booking status.
- To implement a vehicle maintenance tracking module that monitors servicing schedules and vehicle status.

4. Literature Review

- Vehicle rental in many small and mid-sized agencies is still handled through manual registers, telephone bookings, or generic spreadsheet tools.
- These approaches make it difficult to maintain an accurate, real-time record of which vehicles are available, booked, or undergoing maintenance, and they offer little support for generating reports or tracking rental history.
- A number of online rental platforms do exist, but many are built around a single vehicle category or a specific city, and few provide an integrated module for tracking vehicle maintenance alongside bookings.
- This gap between customer-facing booking features and back-end fleet management motivates the design of a single, integrated system for both functions.

5. Proposed Methodology

The proposed system will be developed as a responsive frontend web application using HTML, CSS, and JavaScript.

HTML will be used to create the basic structure of pages such as the home page, vehicle listing, vehicle details, booking page, and admin dashboard.

CSS will be used to create responsive layouts, vehicle cards, forms, tables, navigation components, and dashboard elements.

JavaScript will provide dynamic functionality such as vehicle search and filtering, booking form validation, rental-day calculation, total price calculation, vehicle status updates, and demo booking confirmation.

The system will contain two major interfaces:

- **Customer Interface:**

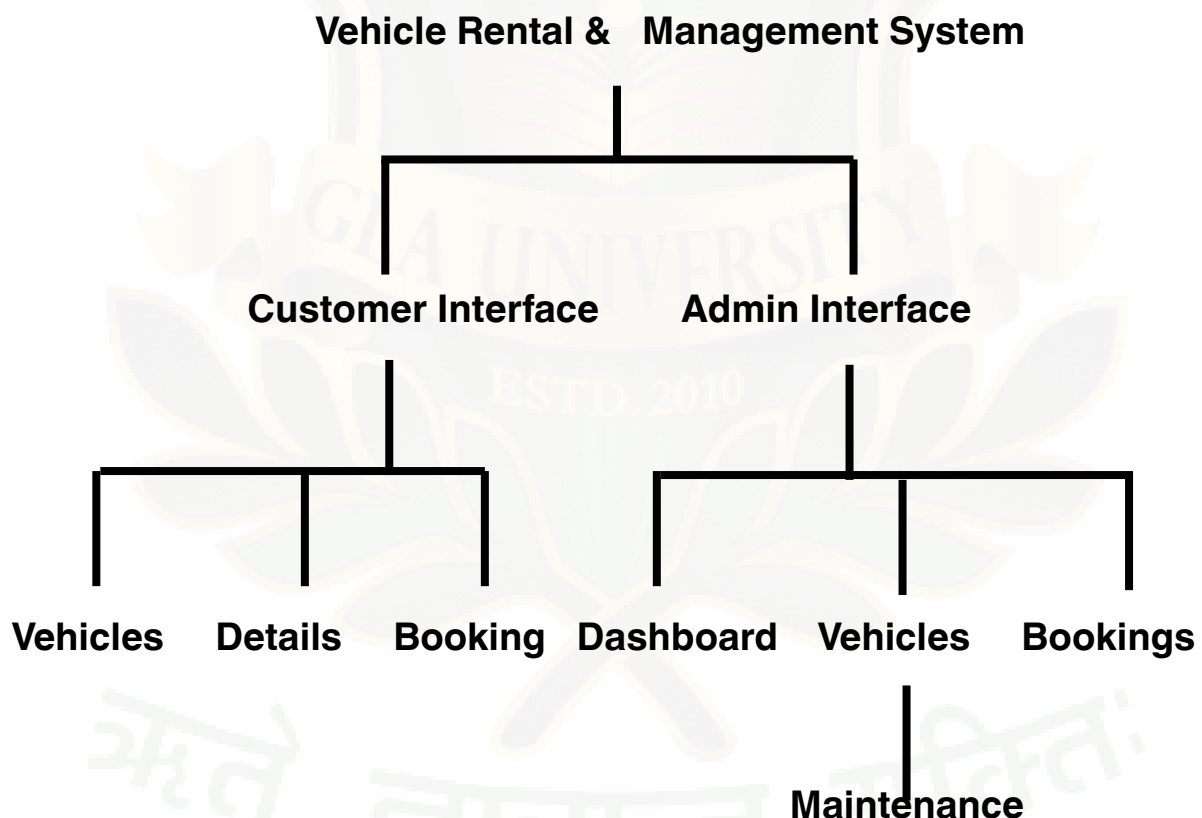
Users can browse vehicles, view details, select rental dates, calculate rental costs, and make demo bookings.

- **Admin Interface:**

The administrator can view dashboard statistics and manage sample vehicle, booking, customer, and maintenance information.

Sample data can be stored in JavaScript objects or LocalStorage to demonstrate frontend functionality without requiring a backend server.

6. System Architecture



The HTML layer will define the structure of the application. The CSS layer will handle styling and responsive layouts. The JavaScript layer will control user interactions and dynamic operations. LocalStorage may be used for temporarily storing demo booking or vehicle information within the browser.

7. Implementation Plan

The implementation will be carried out in different stages.

First, the basic project structure and webpage layouts will be created using HTML. Next, CSS will be used to design the navigation bar, vehicle cards, forms, tables, dashboard, and responsive layouts.

JavaScript will then be integrated to implement vehicle search, filtering, booking calculations, form validation, and interactive elements. Sample vehicle and booking data will be added for demonstration.

The admin dashboard will be developed to display fleet statistics, vehicle information, booking records, and maintenance status. Finally, the complete application will be tested on different screen sizes to ensure proper functionality and responsiveness.

8. Expected Results

The expected result is a responsive and interactive Vehicle Rental & Fleet Management System frontend developed using HTML, CSS, and JavaScript.

The system will allow users to browse and filter vehicles, view vehicle details, select rental dates, calculate rental costs, and make demo bookings. The admin interface will provide an organized dashboard for viewing vehicle, booking, customer, and maintenance information.

The project will demonstrate the practical application of frontend web development concepts and provide a foundation for future integration with backend technologies and databases.

9. Conclusion

The Vehicle Rental & Fleet Management System provides a simple and modern frontend solution for demonstrating vehicle rental and fleet management operations. The system focuses on improving the presentation of vehicle information and providing an interactive rental workflow through a responsive web interface.

Using HTML, CSS, and JavaScript, the project demonstrates important frontend concepts including webpage structure, responsive design, DOM manipulation, form validation, dynamic calculations, filtering, and browser-based data handling.

Although the current project is limited to frontend functionality, it can serve as a foundation for developing a complete vehicle rental system by adding a backend, database, authentication, online payment, and real-time vehicle management in the future.

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